CD36, Sex Differences, Microgravity, and Osteoarthritis

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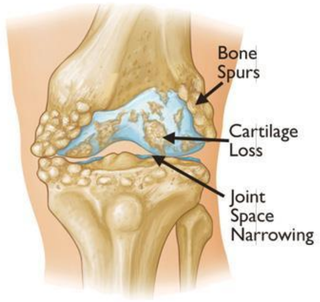
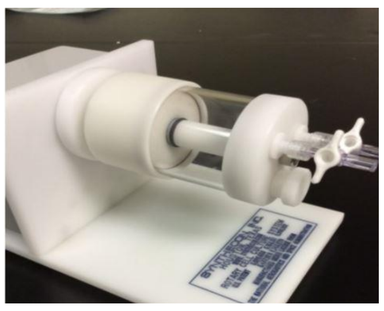
Sex Matters in Knee Osteoarthritis (OA)

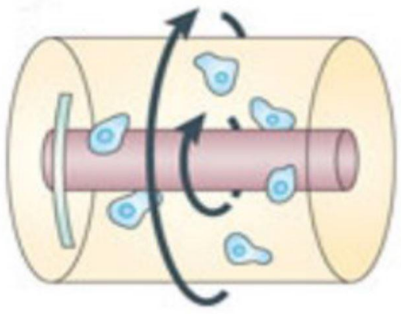
- **Degeneration of the knee joint**
- Affects all structures in the joint
- Develops from little to no mechanical loading of the joint.
- The prevalence is **50% of the population above the age of 65**
- **No current concrete account for large gender disparity**



Knee Osteoarthritis (KOA) incident rate and severity is disproportionately higher in females[2]

Simulated Microgravity (SMG)-RWV Bioreactor





Rotating Wall Vessel (RWV) Bioreactor

- Horizontally rotating cylinder with a coaxial oxygenator.
- Vessel is filled with media and rotates as a **solid body.**
- **Induces MG** on the meniscus samples as there is no relative motion of the samples to the media.
- **Novel platform** for replicating a near zero-gravity environment.
- We would expect this platform to accelerate Osteoarthritic changes.

CD36 and Osteoarthritis

- Prior research suggests that **CD36 may contribute to the inflammatory environment** within the joint.[1]
- CD36-mediated uptake of oxLDL and other ligands can lead to the accumulation of lipid-laden macrophages **(foam cells) in joint tissues.**[3]
- These foam cells release **inflammatory mediators that exacerbate cartilage degradation** and joint inflammation.

Aims

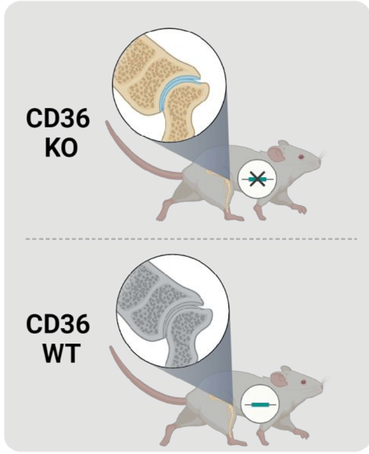
Determine the severity of Osteoarthritis by analyzing cartilage samples through:

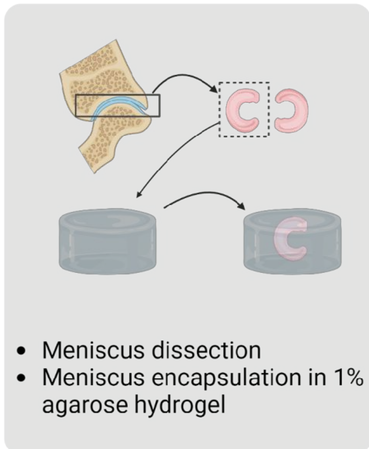
1. Quantifying the **Glycosaminoglycan content** using Safranin-O staining.
2. Examining the content of **Collagen II** using Immunofluorescence staining.
3. Quantifying the degree of **Calcification** using Micro-CT Scans.

Methodology

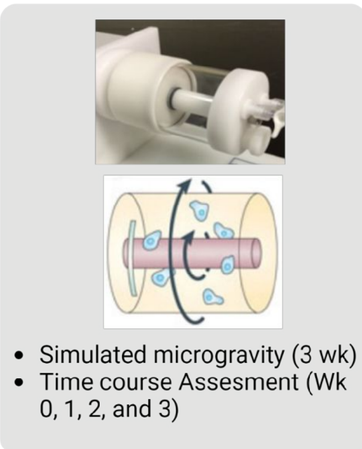
Hypothesis: Long term exposure to SMG will induce an Osteoarthritis-like phenotype in meniscus excised from C57BL/6 mice. This platform can be used to study the impact of CD36, a surface cell marker, in the observed sex-specific variations associated with Knee Osteoarthritis.

Animal Model → Meniscus Encapsulation → Experiment → Analyses

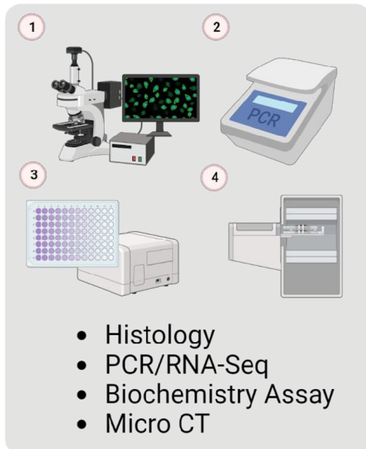




- Meniscus dissection
- Meniscus encapsulation in 1% agarose hydrogel

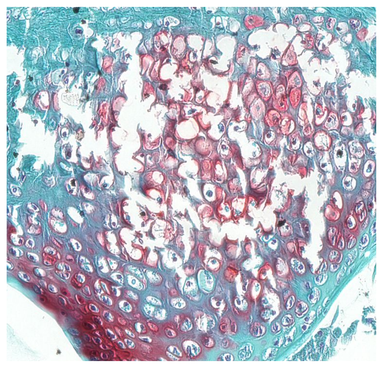


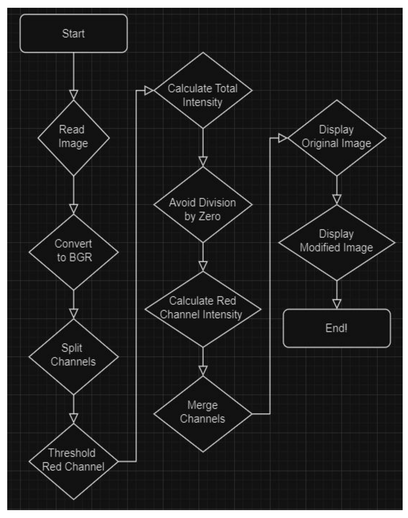
- Simulated microgravity (3 wk)
- Time course Assesment (Wk 0, 1, 2, and 3)



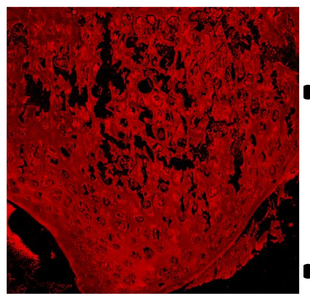
- Histology
- PCR/RNA-Seq
- Biochemistry Assay
- Micro CT

Safranin-O Staining





Quantification with Python



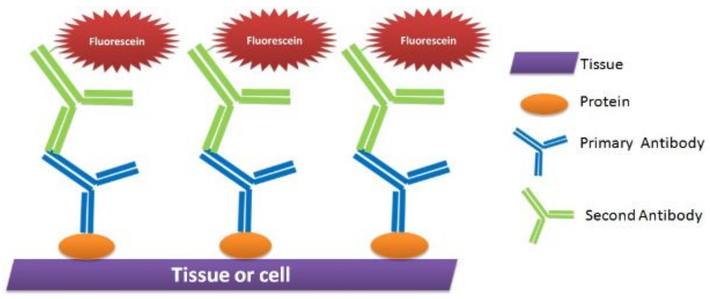
ARIV = 979787.94

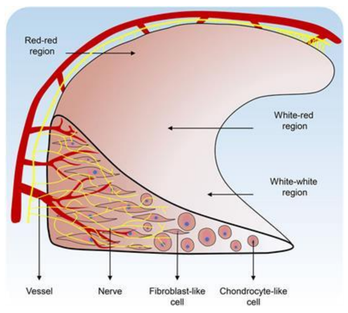
Why Safranin-O?

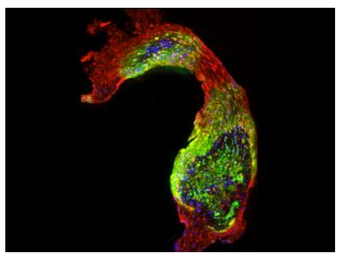
- Utilizing the average red intensity value (ARIV) in biomarker-stained tissues **enables OA assessment.**
- Saf-O stains **GAGs**, which are **responsible for healthy cartilage**

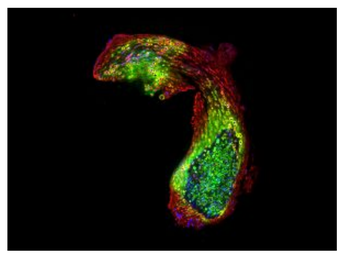
Glycosaminoglycan Content (ARIV) is Inversely Proportional to the Severity of Osteoarthritis

Immunofluorescence Staining









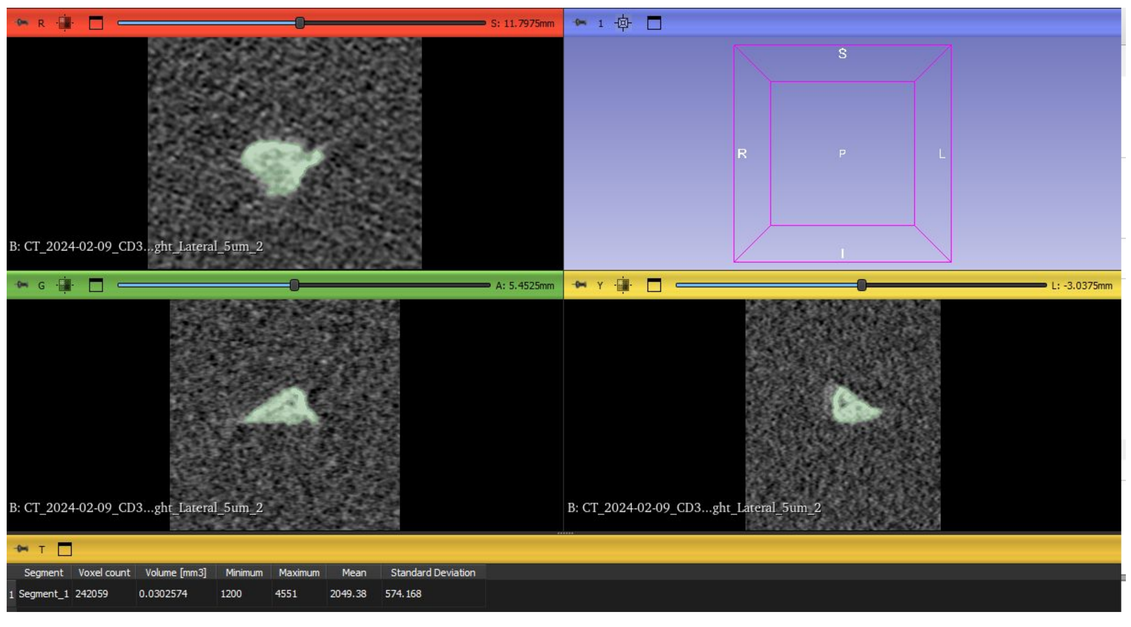
- **Collagen I**
- **Collagen II**
- **Nuclei**

Why Immunofluorescence?

- Examining, the spread and intensity of Collagen II enables **OA assessment**
- Immunofluorescence stains **Collagen I and II**, which are **responsible for healthy cartilage**

Collagen II intensity and distribution in the meniscus is Inversely Proportional to the Severity of Osteoarthritis

uCT Machine Scanning



Why Micro-CT?

- Quantifies the total volume of calcification, which **enables an assessment of OA.**
- 3D Slicer quantifies calcification volume, which is **a clinical marker of Osteoarthritis**

Calcification Volume is Proportional to the Severity of Osteoarthritis

Funding

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UNIVERSITY OF ALBERTA

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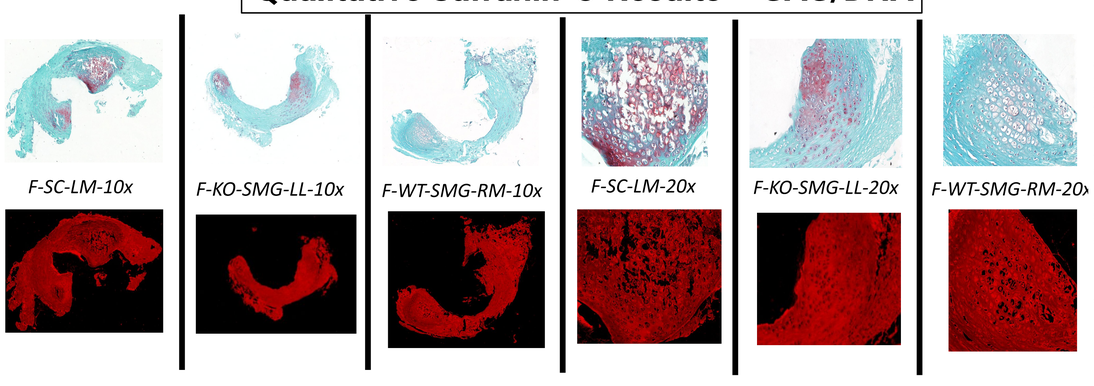
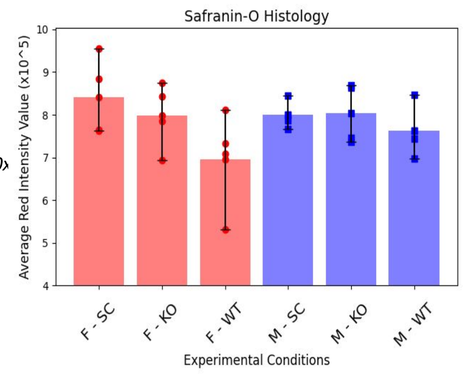
GAG Quantification - Figures

Qualitative Safranin-O Results – GAG/DNA



Quantitative Results

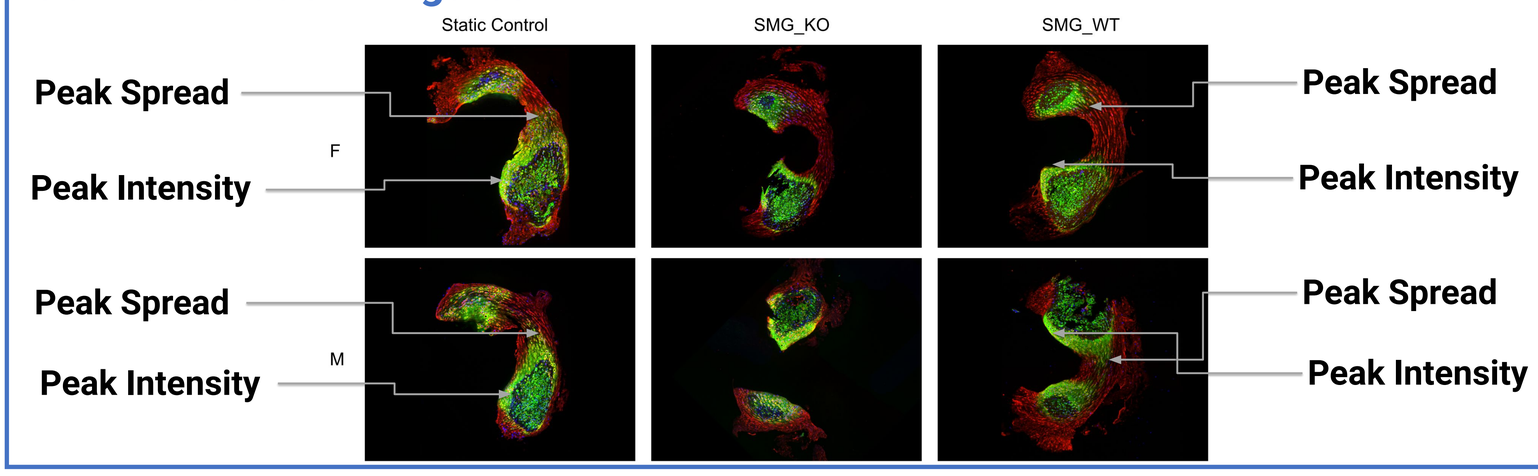


GAG Quantification - Results

- The RVW-SMG Bioreactor **induces a notable decrease in GAG** – namely in females
- The **KO Group in females shows significant increase in GAG content** compared to the SMG group
- Thus, the KO group shows a **protective effect against OA development**

F-SC vs. F-WT	F-WT vs. F-KO	F-KO vs. F-SC
$p < 0.05$	$p < 0.01$	$p > 0.1$

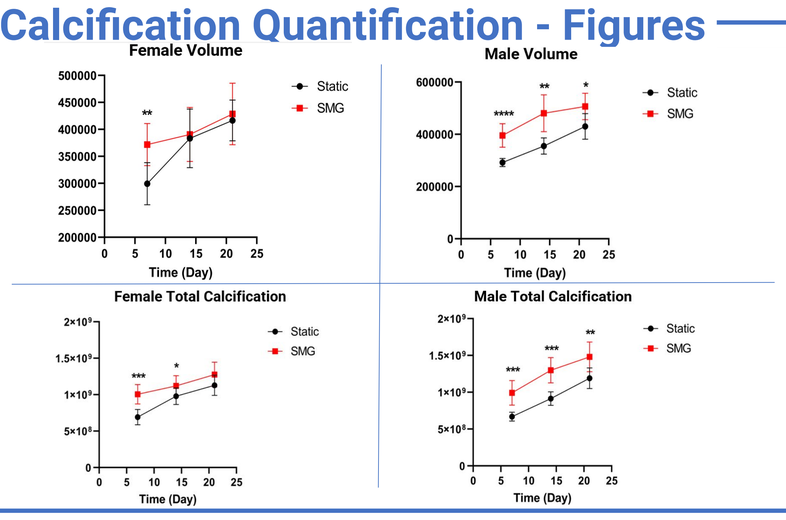
COL2 Evaluation - Figures

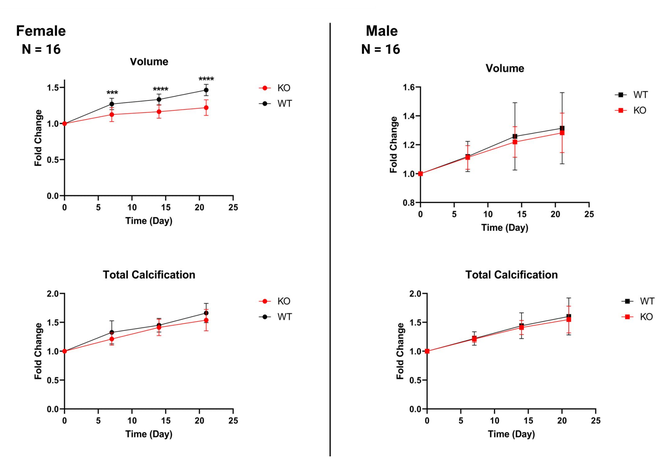


COL2 Evaluation - Conclusion

- The RVW-SMG Bioreactor **induces a notable decrease in Collagen II**, especially in females.
- Smaller area and dimmer intensity of Collagen II was observed in the KO and WT group, **alluding to a more aggressive OA phenotype under simulated microgravity.**
- SC group shows more articular cartilage morphology around the periphery; thus, **SC Group is less hypertrophic compared to the SMG group** based on morphology.

Calcification Quantification - Figures





Calcification Quantification - Conclusion

- The RVW-SMG Bioreactor **induces a notable increase in the volume and intensity of calcification** in both female and male groups compared to the static groups.
- In the SMG reactor, the **KO Group in females shows significant decrease in volume calcified** compared to the WT group, while not a significant difference in total calcification
- **The absence of CD36 keeps calcification localized**, with a decrease in calcification across the tissue

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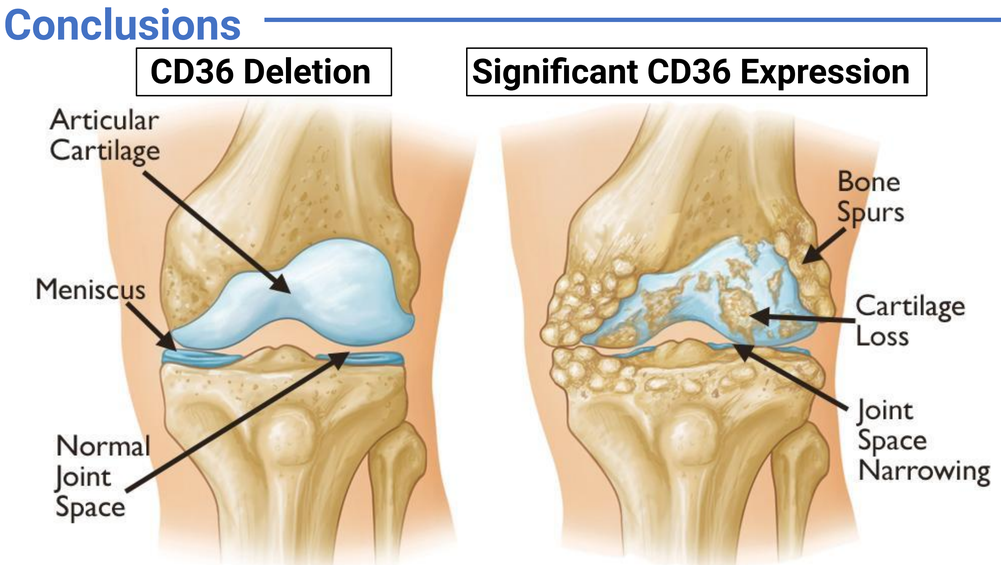
$p < 0.05$	$p < 0.005$	$p < 0.0005$	$p < 0.0001$

Final Conclusions



- **Exposure to Simulated Microgravity** using the Rotating Wall Vessel Bioreactor **resulted in increased Osteoarthritic phenotypes as per the most clinical test (uCT), in both** KO and WT groups, establishing the **SMG-RWV Bioreactor as a satisfactory platform.**
- The **CD36 deletion significantly mitigated Osteoarthritic phenotypes within females**, but not males, with the KO group exhibiting a similar phenotype to the SC group.

Real-World Implications

- **Flow Cytometry –** by examining the expression of CD36 in other tissues, **Osteoarthritis can be screened for.**
- **Spaceflight Screening – Astronauts can be screened for spaceflight** with respect to the expression of CD36.
- **CD36 Inhibition –** Due to its proven role in OA development, future drugs could be manufactured **to inhibit the expression of CD36.**

Thank You

Firstly, I'd like to thank **Melanie Kunze, Aillette Mulet Sierra, and Dr. Adesida** for welcoming me to **Adesida Laboratory**, and for teaching me the procedures of the lab. I'd also like to especially thank **Zhiyao (Hilda) Ma** for taking the time to supervise me, and for being there whenever I had questions. Overall, she was an amazing mentor. Lastly, I'd like to thank my **friends, family**, and the **ERSF committee** for the support, feedback, motivation they have given me over the past few months.

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